



Mario Lavanga

Senior Research Engineer | Health AI & Causal Inference
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EDUCATION

KU LEUVEN

PHD IN ELECTRICAL ENGINEERING
Grad. September 2020 | Leuven,
Belgium | Phd Graduation Video

POLITECNICO DI MILANO

MS IN BIOMEDICAL ENGINEERING
Grad. December 2015 | Milan, Italy |
Final Grade: 100/100 cum laude

BS IN BIOMEDICAL ENGINEERING

Grad. December 2015 | Milan, Italy |
Final Grade: 100/100 cum laude

ALTA SCUOLA POLITECNICA

DOUBLE DEGREE WITH POLITECNICO
DI TORINO
Grad. December 2015 | Turin, Italy

LINKS

Mario Lavanga
 Mario Lavanga
 Mario Lavanga

SKILLS

PROGRAMMING

Python • Matlab • C# • Bash • Powershell

FRAMEWORKS

Azure • Azure AI Foundry • Azure DevOps •
Tensorflow • Git

STATISTICS & MODELLING

Causal Inference (Bayesian, Granger,
PCMCi) • Time-Series & Longitudinal
Modelling • Uncertainty Quantification
(VAE, SNPE) • Hypothesis Testing

LANGUAGES

Italian | Mother tongue • English, German,
French | Fluent • 汉语, Dutch | Basic

AI CODING TOOLS

Claude Code • Codex • Cline • Gemini CLI •
MCP integrations

ABOUT ME

I build deep-learning and signal-processing models on physiological data, then prove they hold up on real hospital data before they reach a patient. My methods: causal inference (PCMCi, Granger, Bayesian), time-series modelling and self-supervised representation learning, and uncertainty quantification, on observational ICU and NICU data under FDA constraints. My background spans the aging brain, real-time ICU monitoring, and neonatal stress detection. Offstage I act in amateur theatre and keep collecting languages.

RESEARCH

HAMILTON MEDICAL AG | PROJECT LEADER

December 2024 –Present | Domat/Ems, Switzerland

Leading cloud ventilation-management features from research through IP protection and regulatory submission.

HAMILTON MEDICAL AG | SENIOR RESEARCH ENGINEER

June 2022 –Present | Domat/Ems, Switzerland

Built deep-learning and rule-based models for patient-ventilator asynchrony, respiratory mechanics, and patient effort, and shipped the asynchrony detectors both on the ventilator itself (embedded, real-time) and in the cloud platform. Applied causal discovery (PCMCi / Granger) on ICU time series to find asynchrony drivers. Working on patent applications.

TNG - INS | POSTDOCTORAL RESEARCHER

Oct 2020 –June 2022 | Marseille, France

Collaborated with Prof. Viktor Jirsa and Prof. Svenja Caspers to investigate virtual brain aging (Digital Twin) using causal inference and self-supervised representation learning with uncertainty quantification (Variational Autoencoders) on anatomical data. Part of the Human Brain Project, showcased on EBRAINS (code).

BIOMED GROUP - STADIUS | PHD FELLOW

Jan 2016 –Sep 2020 | Leuven, Belgium

Developed a Perinatal Stress Calculator using multimodal biosignals (ECG, HRV, EEG) in collaboration with Prof. Sabine Van Huffel and Prof. Gunnar Naulaers. Focused on data-driven stress quantification in premature babies via longitudinal analysis and robust features in noisy clinical environments. Collaboration with UZ Leuven.

SELECTED PUBLICATIONS

M. Lavanga et al., The virtual aging brain: Causal inference supports interhemispheric dedifferentiation in healthy aging. *NeuroImage*, 2023.
M. Lavanga et al., The effect of early procedural pain in preterm infants on the maturation of EEG and heart rate variability. *Pain*, 2021.
Full list on Google Scholar —611 citations, h-index 13, i10 18.

AWARDS AND VARIA

- Guest Editor for *Frontiers in Network Physiology* (2021-2022) | Talk at University of Virginia hosted by Prof. Randall Moorman.
- Falling Walls Lab Leuven Participant (2019) | KU Leuven Tech Transfer Laureate.
- FWO Strategic Basic Research Fellowship Recipient (2016).